

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

QUESTION BANK

Regulation	:	2023
Course Name	:	DATABASE MANAGEMENT SYSTEMS
Course Code	:	JCS2401
Year / Semester / Branch	:	II / IV / CSE
Unit No	:	5
Unit Title	:	ADVANCED DATABASES
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

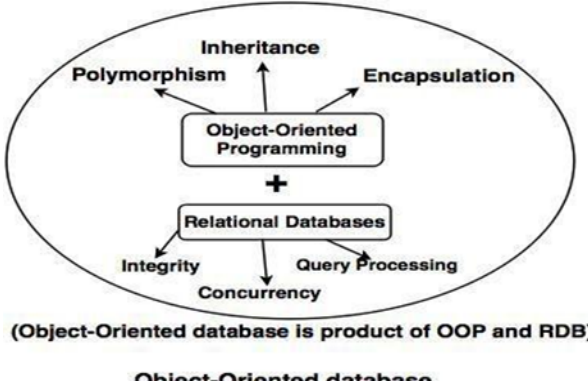
Q.No.	Questions	Marks	BTL	CO
1	Define Object-Oriented Database (OODBMS). Answer: An Object-Oriented Database Management System (OODBMS) is a database system that stores, manages, and retrieves data in the form of objects, as used in object-oriented programming (OOP). It integrates database capabilities with object-oriented programming features such as classes, objects, inheritance, polymorphism, and encapsulation.	2	K1	CO5
2	List the advantages of OODB ● Supports complex data types (e.g., multimedia, images, videos). ● Encapsulation improves data security and integrity. ● Inheritance enables reuse of data and reduces redundancy. ● Better modeling of real-world entities using objects. ● Seamless integration with object-oriented programming languages. ● Reduced impedance mismatch between application and database.	2	K1	CO5
3	Outline the aspects of Objects and Literals. Objects are entities that have identity, state (attributes), and behavior (methods). Literals are constant values that do not have identity and are	2	K2	CO5



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	used to represent data directly.															
4	Define ODL. ODL language is used to create object specifications: classes and interfaces-Using the specific language bindings to specify how ODL constructs can be mapped to constructs in specific programming language, such as C++, SMALLTALK, and JAVA	2	K1	CO5												
5	Give the diagrammatic representation Object Oriented Database. 	2	K1	CO5												
6	How does an Object-relational database work? An Object-Relational Database combines relational tables with object-oriented features, allowing data to be stored in tables along with complex data types like objects, arrays, and user-defined types. It works by extending SQL to support object concepts such as inheritance, methods, and encapsulation, enabling more flexible and powerful data modeling.	2	K2	CO5												
7	Differentiate between E-R and ODL <table><tr><th>Feature</th><th>E/R Model (Entity-Relationship)</th><th>ODL (Object Definition Language)</th></tr><tr><td>Approach</td><td>Conceptual data model</td><td>Object-oriented data model</td></tr><tr><td>Representation</td><td>Uses entities, attributes, relationships</td><td>Uses objects, classes, and methods</td></tr><tr><td>Data Types</td><td>Simple data types</td><td>Supports complex and user-defined data types</td></tr></table>	Feature	E/R Model (Entity-Relationship)	ODL (Object Definition Language)	Approach	Conceptual data model	Object-oriented data model	Representation	Uses entities, attributes, relationships	Uses objects, classes, and methods	Data Types	Simple data types	Supports complex and user-defined data types	2	K2	CO5
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	Behaviour	Does not support methods	Supports methods (behaviour)			
	Inheritance	Not supported	Supported			
	Usage	Used for database design (conceptual level)	Used to define object database schema			
8	Mention the functionality of a XML database. An XML database is a data persistence software system that allows data to be stored in XML format. These data can then be queried, exported and serialized into the desired format. XML databases are usually associated with document-oriented databases			2	K1	CO5
9	Give the Syntax for a basic OQL command. class Person (extent persons key ssn) { } class Faculty extends Person ENTRY POINTS (extent faculty) { } class Department (extent department key dname) { }			2	K1	CO5
10	Define the hierarchical structure of XML document XML documents have a hierarchical structure and can conceptually be interpreted as a tree structure, called an XML tree. XML documents must contain a root element (one that is the parent of all other elements). All elements in an XML document can contain sub elements, text and attributes.			2	K1	CO5
11	Interpret the use of XML Schema XML Schema is commonly known as XML Schema Definition (XSD). It is used to describe and validate the structure and the content of XML data. XML schema defines the elements, attributes and data types. Schema element supports Namespaces.			2	K2	CO5
12	List the components of mobile database. • Mobile Client • Wireless Communication Network • Server (Fixed Host) • Database Management System (DBMS) • Middleware			2	K1	CO5



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13	Differentiate between OODBMS and RDBMS.			2	K2	CO5
	Feature	OODBMS	RDBMS			
	Data Model	Object-oriented (objects & classes)	Relational (tables & rows)			
	Data Representation	Stores data as objects	Stores data in tables			
	Inheritance	Supported	Not supported			
	Query Language	Uses OQL or similar	Uses SQL			
14	Illustrate on DTD. A document type definition (DTD) contains a set of rules that can be used to validate an XML file. After you have created a DTD, you can edit it manually, adding declarations that define elements, attributes, entities, and notations, and how they can be used for any XML files that reference the DTD file			2	K2	CO5
15	Outline the steps to create DTD. 1. Create a project to contain the DTD if needed. 2. In the workbench, click File > New > Other and select XML>DTD. Click Next. 3. Select the project or folder that will contain the DTD. 4. In the File name field, type the name of the DTD, for example MyDTD.dtd. The name of your DTD file must end with the extension.dtd .Click Next 5. Optional: You can use a DTD template as the basis for your new DTD file. To do so, click the Use DTD Template check box, and select the template you want to use. 6. Click Finish.			2	K2	CO5
16	Give reasons for why a XML Schema is better considered to a DTD. • It allows the text that appears in elements to be constrained to specific types, such as numeric types in specific formats or complex types such as sequences of elements of other types. • It allows user-defined types to be created. • It allows uniqueness and foreign-key constraints. • It standardizes object database models and query languages for interoperability.			2	K2	CO5
17	How do mobile databases differ from traditional databases? A mobile database is a database that is designed to operate on mobile devices such as smartphones, tablets, and			2	K2	CO5



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	laptops, allowing offline access and synchronization with a central database. It uses centralized and require a constant network connection, mobile databases support disconnected operations, replication, and synchronization for data consistency.			
18	Distinguish between XML and HTML. XML is used for data storage, while HTML is used for data presentation. XML is Case-sensitive (e.g., <Name> and <name> are different), while HTML is not case-sensitive (e.g., <P> is the same as <p>).	2	K2	CO5
19	Illustrate on temporal database. A temporal database in DBMS is a database system that stores and manages data with a time dimension, enabling the tracking of data changes over time and providing historical analysis capabilities.	2	K2	CO5
20	Why is Spatial Index important? A spatial index is used by a spatial database to optimize spatial queries. Database systems use indices to quickly look up values by sorting data values in a linear (e.g. alphabetical) order. Instead, spatial databases use a <i>spatial</i> index designed specifically for multi-dimensional ordering.	2	K2	CO5

PART B

Q.No.	Questions	Marks	BTL	CO
1	Apply the concept of Object-Oriented Data Model to design a database schema for any system of your choice .	16	K3	CO5
2	Differentiate between Object-Oriented Databases and Object-Relational Databases with suitable examples.	16	K3	CO5
3	Illustrate how an ODMG Object Model is created and its role in standardizing Object-Oriented Databases.	16	K2	CO5
4	Construct an XML document for a student database and demonstrate its structure using elements and attributes.	16	K3	CO5
5	Illustrate the process of creating a XML schema for a given database.	16	K2	CO5
6	Summarize the challenges and advantages of Mobile Databases in distributed environments.	16	K2	CO5
7	Demonstrate how Temporal Database model can be used to manage employee records with time-varying data.	16	K3	CO5
8	Discuss in detail about Spatial Database system for a GIS application.	16	K2	CO5

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